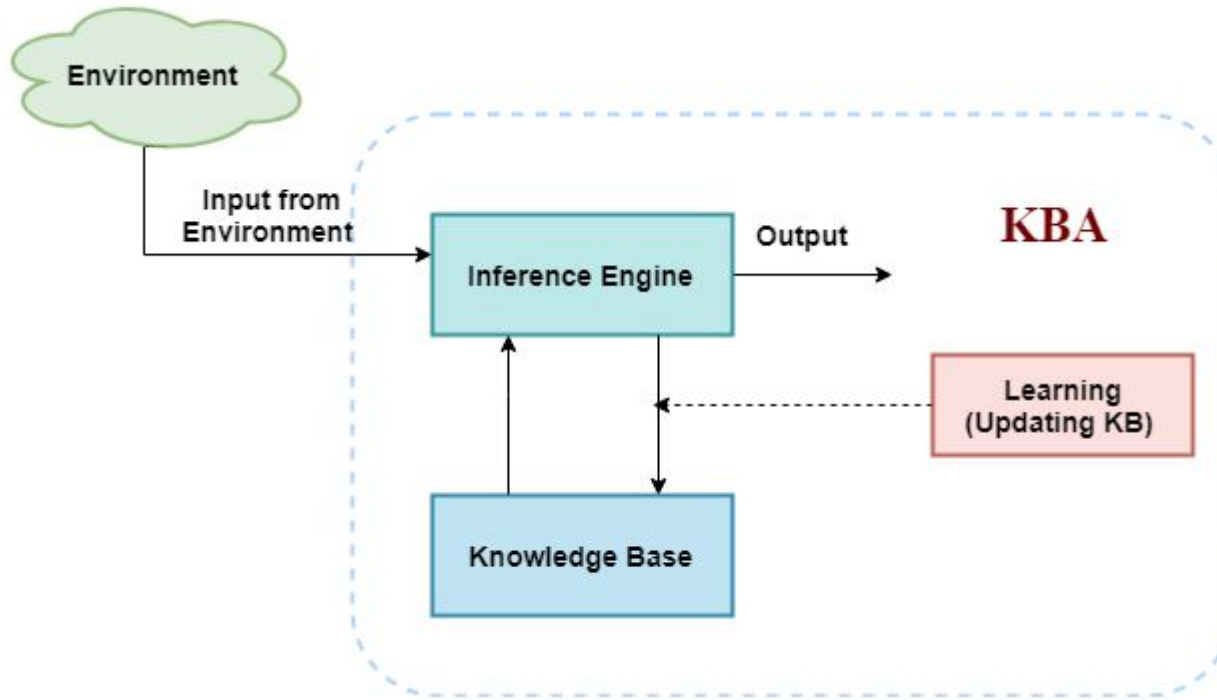


Knowledge Representation

Knowledge-Based Agent

- Knowledge-based agents are those agents who have the capability of maintaining an internal state of knowledge, reason over that knowledge, update their knowledge after observations and take actions. These agents can represent the world with some formal representation and act intelligently.
- Knowledge-based agents are composed of two main parts:
 - Knowledge-base and
 - Inference system.

Architecture of knowledge-based agent:



COMPONENTS

- Knowledge base: Knowledge-base is a central component of a knowledge-based agent, it is also known as KB. It is a collection of sentences (here 'sentence' is a technical term and it is not identical to sentence in English).
- Inference system
- Inference means deriving new sentences from old. Inference system allows us to add a new sentence to the knowledge base. A sentence is a proposition about the world. Inference system applies logical rules to the KB to deduce new information.
- Forward chaining
- Backward chaining

Operations Performed by KBA

- Following are three operations which are performed by KBA in order to show the intelligent behavior:
- TELL: This operation tells the knowledge base what it perceives from the environment.
- ASK: This operation asks the knowledge base what action it should perform.
- Perform: It performs the selected action.

Various levels of knowledge-based agent:

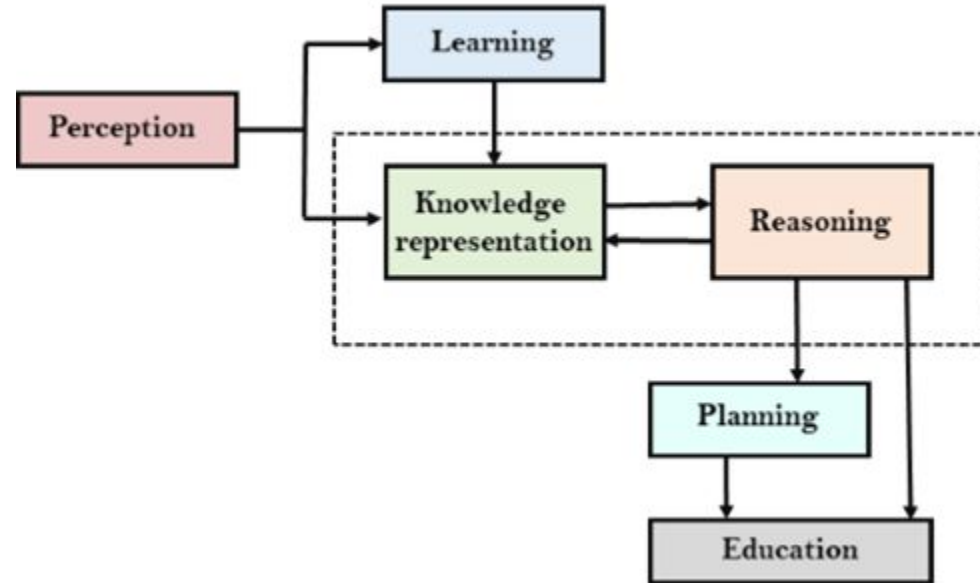
- 1. Knowledge level
- Knowledge level is the first level of knowledge-based agent, and in this level, we need to specify what the agent knows, and what the agent goals are.
- 2. Logical level:
 - At this level, we understand that how the knowledge representation of knowledge is stored. At this level, sentences are encoded into different logics.
- 3. Implementation level:
 - This is the physical representation of logic and knowledge. At the implementation level agent perform actions as per logical and knowledge level.

Knowledge representation

- Knowledge representation and reasoning (KR, KRR) is the part of Artificial intelligence which concerned with AI agents thinking and how thinking contributes to intelligent behavior of agents.
- Types of knowledge



AI knowledge cycle



Approaches to KR

1. Simple relational knowledge:

- Provides very weak inferential capabilities.
- May serve as the input to powerful inference engines.

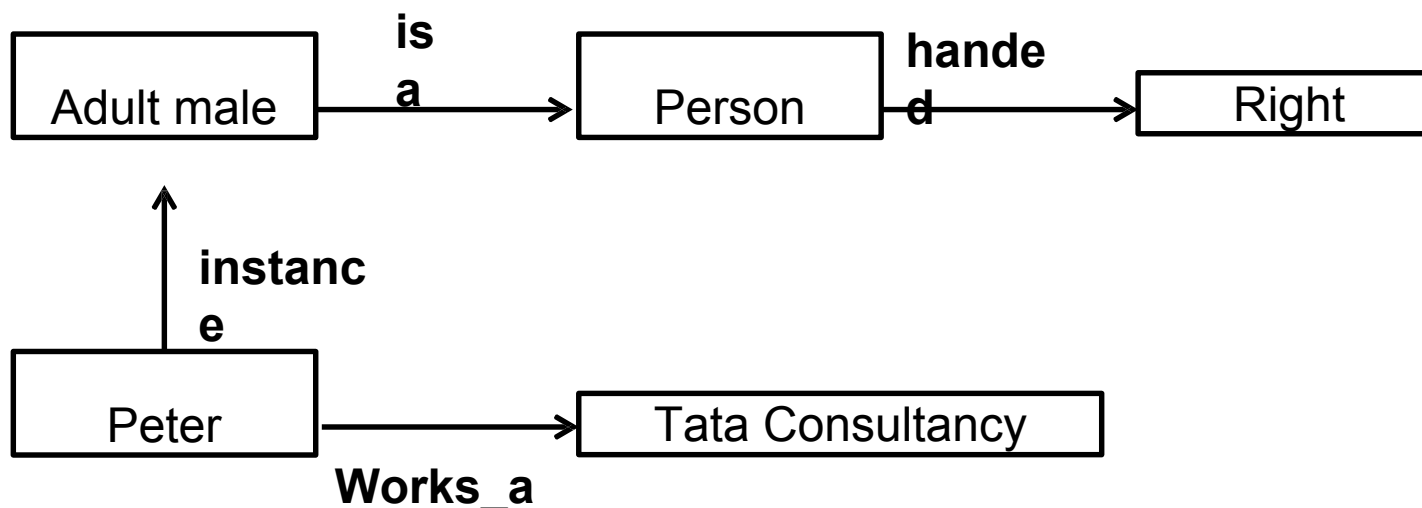
Player	Height	Weight	handed
Peter	6-0	180	right
Ajay	5-10	170	left
John	6-2	215	left
Vickey	6-3	205	right

Fails to infer “which right handed player can best face a particular bowler” .

Approaches to KR

Inheritable knowledge:

- Objects are organized into **classes** and classes are organized in a generalization **hierarchy**.
- **Inheritance** is a powerful form of inference, but **not adequate**.
- Ex. Property inheritance inference mechanism.



Approaches to KR

Inferential knowledge:

- Facts represented in a **logical form**, which facilitates **reasoning**.
- An **inference engine** is required.

ex. 1. “Marcus is a **man**”

2. “All **men** are **mortal**”

Implies:

3. “**Marcus** is **mortal**”

Approaches to KR

Procedural knowledge:

- Representation of “how to make it” rather than “what it is”.
- May have inferential efficiency, but no inferential adequacy and acquisitional efficiency.
- Ex. Writing LISP programs

Techniques used for
Knowledge
Representation

Propositional logic

First order logic

Rule-based System

Semantic Networks

Frames

Script

Techniques of knowledge representation

- **Propositional Logic:** This technique is also known as propositional calculus, statement logic, or sentential logic. It is used for representing the knowledge about what is true and what is false.
- **First-order Logic:** It is also known as Predicate logic or First-order predicate calculus (FOPL). This technique is used to represent the objects in the form of predicates or quantifiers. It is different from Propositional logic as it removes the complexity of the sentence represented by it.
- **Rule-based System:** This is the most commonly used technique in artificial intelligence. In the rule-based system, we impose rules over the propositional logic and first-order logic techniques. If-then clause is used for this technique. For example, if there are two variables A and B. Value of both A and B is True. Consequently, the result of both should also be True and vice-versa. It is represented as:
 - If the value of A and B is True, then the result will be True. So, such a technique makes the propositional as well as FOPL logics bounded in the rules.

- **Semantic Networks:** The technique is based on storing the knowledge into the system in the form of a graph. Nodes of a graph represent the objects which exist in the real world, and the arrow represents the relationship between these objects. Such techniques show the connectivity of one object with another object.

Ram has a cycle.

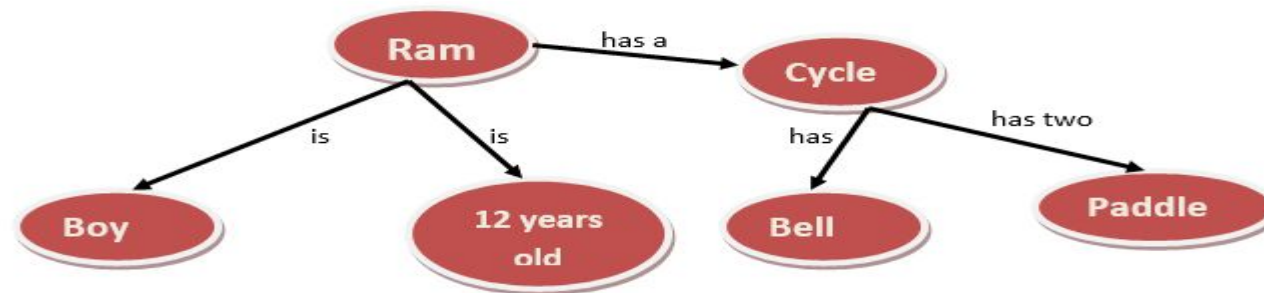
Ram is a boy.

Cycle has a bell.

Ram is 12 years old.

Cycle has two paddles.

Representing the above information in the form of a graph looks like:



- Frames: In this technique, the knowledge is stored via slots and fillers. Similarly, the Slots are the entities and Fillers are its attributes. They are together stored in a frame. So, whenever there is a requirement, the machine infers the necessary information to take the decision. For example, Tomy is a dog having one tail.
- Script: It is an advanced technique over the Frames. Here, the information is stored in the form of a script. The script is stored in the system containing all the required information.

Propositional logic

- Statements used in mathematics.
- Proposition :is a declarative sentence whose value is either true or false.
- Examples:
 - “The sky is blue.” [Atomic Proposition]
 - “The sky is blue and the plants are green.”
[Molecular/Complex Proposition]
 - “Today is a rainy day” [Atomic Proposition]
 - “Today is Sunday” [Atomic Proposition]
 - “ $2*2=4$ ” [Atomic Proposition]

RULES

- Propositional logic is also called Boolean logic as it works on 0 and 1.
- In propositional logic, we use symbolic variables to represent the logic, and we can use any symbol for a representing a proposition, such A, B, C, P, Q, R, etc.
- Propositions can be either true or false, but it cannot be both.
- Propositional logic consists of an object, relations or function, and **logical connectives**.
- These connectives are also called logical operators.
- The propositions and connectives are the basic elements of the propositional logic.
- Connectives can be said as a logical operator which connects two sentences.
- A proposition formula which is always true is called **tautology**, and it is also called a valid sentence.
- A proposition formula which is always false is called **Contradiction**.
- Statements which are questions, commands, or opinions are not propositions such as "**Where is Rohini**", "**How are you**", "**What is your name**", are not propositions.

Logical connectives

- Logical connectives are used to connect two simpler propositions or representing a sentence logically.
- There are mainly five connectives, which are given as follows:
 - **Negation:** A sentence such as $\neg P$ is called negation of P. A literal can be either Positive literal or negative literal.
 - **Conjunction:** A sentence which has $\wedge$ connective such as, $P \wedge Q$ is called a conjunction.
Example: Rohan is intelligent and hardworking. It can be written as,
P= Rohan is intelligent,
Q= Rohan is hardworking. $\rightarrow P \wedge Q$.

- Disjunction: A sentence which has $\vee$ connective, such as $P \vee Q$. is called disjunction, where P and Q are the propositions.
Example: "Ritika is a doctor or Engineer",
Here P= Ritika is Doctor. Q= Ritika is Doctor, so we can write it as $P \vee Q$.
- Implication: A sentence such as $P \rightarrow Q$, is called an implication. Implications are also known as if-then rules. It can be represented as
If it is raining, then the street is wet.
Let P= It is raining, and Q= Street is wet, so it is represented as $P \rightarrow Q$
- Biconditional: A sentence such as $P \Leftrightarrow Q$ is a Biconditional sentence, example If I am breathing, then I am alive
P= I am breathing, Q= I am alive, it can be represented as $P \Leftrightarrow Q$.

Propositional Logic Connectives:

Connective symbols	Word	Technical term	Example
$\wedge$	AND	Conjunction	$A \wedge B$
$\vee$	OR	Disjunction	$A \vee B$
$\rightarrow$	Implies	Implication	$A \rightarrow B$
$\leftrightarrow$	If and only if	Biconditional	$A \leftrightarrow B$
$\neg$ or $\sim$	Not	Negation	$\neg A$ or $\neg B$

For Negation:

P	$\neg P$
True	False
False	True

For Conjunction:

P	Q	$P \wedge Q$
True	True	True
True	False	False
False	True	False
False	False	False

For disjunction:

P	Q	$P \vee Q$
True	True	True
False	True	True
True	False	True
False	False	False

For Implication:

P	Q	$P \rightarrow Q$
True	True	True
True	False	False
False	True	True
False	False	True

For Biconditional:

P	Q	$P \leftrightarrow Q$
True	True	True
True	False	False
False	True	False
False	False	True

We can build a proposition composing three propositions P, Q, and R. This truth table is made-up of 8n Tuples as we have taken three proposition symbols.

P	Q	R	$\neg R$	$P \vee Q$	$P \vee Q \rightarrow \neg R$
True	True	True	False	True	False
True	True	False	True	True	True
True	False	True	False	True	False
True	False	False	True	True	True
False	True	True	False	True	False
False	True	False	True	True	True
False	False	True	False	False	True
False	False	False	True	False	True

Precedence	Operators
First Precedence	Parenthesis
Second Precedence	Negation
Third Precedence	Conjunction(AND)
Fourth Precedence	Disjunction(OR)
Fifth Precedence	Implication
Six Precedence	Biconditional

Logical equivalence:

- Logical equivalence is one of the features of propositional logic. Two propositions are said to be logically equivalent if and only if the columns in the truth table are identical to each other.
- Let's take two propositions A and B, so for logical equivalence, we can write it as $A \Leftrightarrow B$. In below truth table we can see that column for $\neg A \vee B$ and $A \rightarrow B$, are identical hence A is Equivalent to B

A	B	$\neg A$	$\neg A \vee B$	$A \rightarrow B$
T	T	F	T	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

INFERENCE RULES IN PROPOSITIONAL LOGIC

- **Commutativity:**

- $P \wedge Q = Q \wedge P$, or
- $P \vee Q = Q \vee P$.

- **Associativity:**

- $(P \wedge Q) \wedge R = P \wedge (Q \wedge R)$,
- $(P \vee Q) \vee R = P \vee (Q \vee R)$

- **Identity element:**

- $P \wedge \text{True} = P$,
- $P \vee \text{True} = \text{True}$

INFERENCE RULES IN PROPOSITIONAL LOGIC

- **Distributive:**

- $P \wedge (Q \vee R) = (P \wedge Q) \vee (P \wedge R).$
- $P \vee (Q \wedge R) = (P \vee Q) \wedge (P \vee R).$

- **DE Morgan's Law:**

- $\neg (P \wedge Q) = (\neg P) \vee (\neg Q)$
- $\neg (P \vee Q) = (\neg P) \wedge (\neg Q).$

- **Double-negation elimination:**

- $\neg (\neg P) = P.$

Inference rules:

- Following are some terminologies related to inference rules:
- **Implication:** It is one of the logical connectives which can be represented as $P \rightarrow Q$. It is a Boolean expression.
- **Converse:** The converse of implication, which means the right-hand side proposition goes to the left-hand side and vice-versa. It can be written as $Q \rightarrow P$.
- **Contrapositive:** The negation of converse is termed as contrapositive, and it can be represented as $\neg Q \rightarrow \neg P$.
- **Inverse:** The negation of implication is called inverse. It can be represented as $\neg P \rightarrow \neg Q$.

P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$\neg Q \rightarrow \neg P$	$\neg P \rightarrow \neg Q$
T	T	T	T	T	T
T	F	F	T	F	T
F	T	T	F	T	F
F	F	T	T	T	T

RULES

Modus Ponens:

If **P** is true and **P→Q** then we can infer **Q** is also true.

$$\begin{array}{c} \mathbf{P} \\ \mathbf{P \rightarrow Q} \\ \hline \mathbf{Hence, Q} \end{array}$$

Example:

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I am sleepy" $\implies P$

Conclusion: "I go to bed." $\implies Q$.

Hence, we can say that, if $P \rightarrow Q$ is true and P is true then Q will be true.

RULES

- Modus Tollens

If $\neg P$ is true and $P \rightarrow Q$ then we can infer $\neg Q$.

$$\begin{array}{c} \neg P \\ P \rightarrow Q \\ \hline \text{Hence, } \neg Q \end{array}$$

Statement-1: "If I am sleepy then I go to bed" $\implies P \rightarrow Q$

Statement-2: "I do not go to the bed." $\implies \neg Q$

Statement-3: Which infers that "I am not sleepy" $\implies \neg P$

RULES

Hypothetical Syllogism

The Hypothetical Syllogism rule state that if $P \rightarrow R$ is true whenever $P \rightarrow Q$ is true, and $Q \rightarrow R$ is true. It can be represented as the following notation:

Example:

Statement-1: If you have my home key then you can unlock my home. $P \rightarrow Q$

Statement-2: If you can unlock my home then you can take my money. $Q \rightarrow R$

Conclusion: If you have my home key then you can take my money. $P \rightarrow R$

Disjunctive Syllogism:

if $\neg p$ and $p \vee q$ we can infer q is true.

Example:

Statement-1: Today is Sunday or Monday. $\Rightarrow P \vee Q$

Statement-2: Today is not Sunday. $\Rightarrow \neg P$

Conclusion: Today is Monday. $\Rightarrow Q$

RULES

Addition:

It states that If P is true, then $P \vee Q$ will be true.

Example:

Statement: I have a vanilla ice-cream. $\implies P$

Statement-2: I have Chocolate ice-cream.

Conclusion: I have vanilla or chocolate ice-cream. $\implies (P \vee Q)$

Simplification:

If $P \wedge Q$ is true, then **Q or P** will also be true. It can be represented as:

Resolution:

If $P \vee Q$ and $\neg P \wedge R$ is true, then $Q \vee R$ will also be true. **It can be represented as**

Prove the following using either resolution or traditional logic, using these propositions:

S : I study; G: I get good grades; E: I enjoy.

1. If I study I make good grades.
2. If I do not study I enjoy.
 $\therefore$ either I make good grades or I enjoy.

Solution:

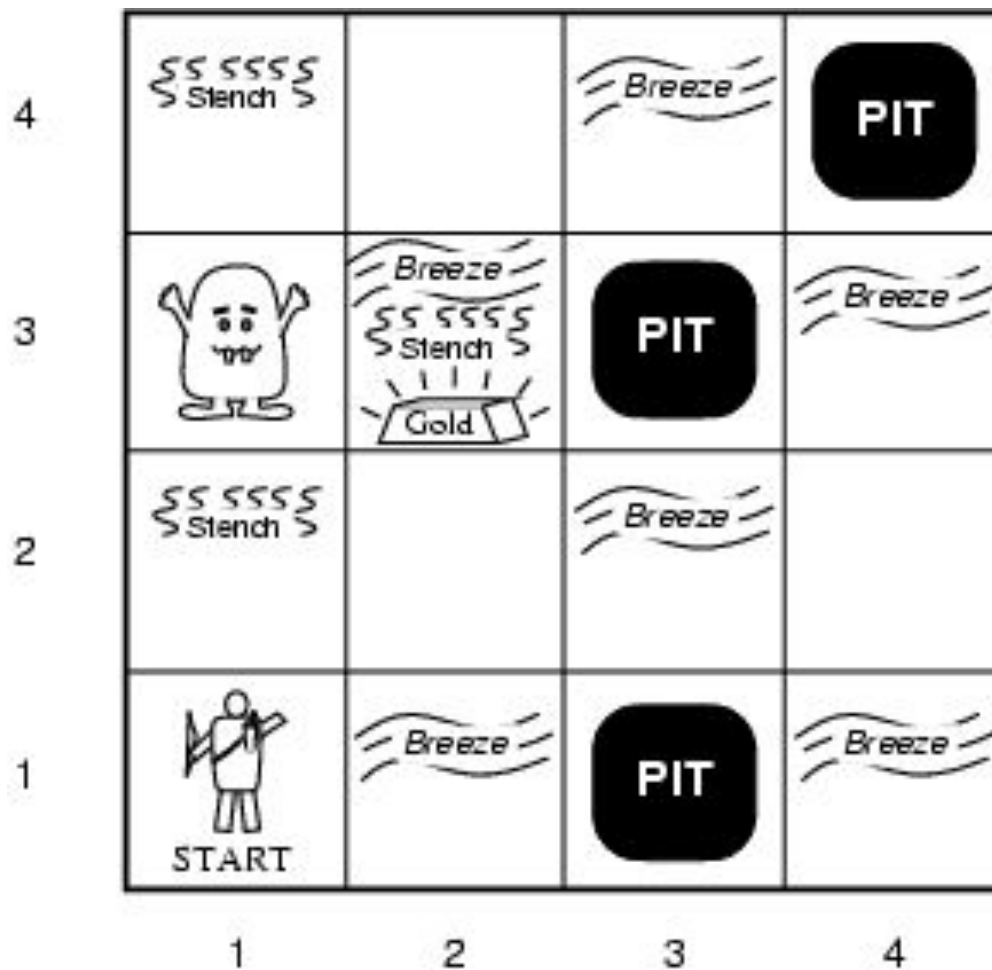
Traditional logic proof (one of many possible solutions):

1. $S \Rightarrow G$
2. $\neg S \Rightarrow E$
 $\therefore G \vee E$
3. $\neg E \Rightarrow S$ (2; transp.+D.N.)
4. $\neg E \Rightarrow G$ (3,1; H.S.)
5. $E \vee G$ (4; M.I.+D.N.)
6. $G \vee E$ (5; Comm.) $\square$

Limitations of Propositional logic:

- We cannot represent relations like ALL, some, or none with propositional logic. Example:
 - **All the girls are intelligent.**
 - **Some apples are sweet.**
- Propositional logic has limited expressive power.
- In propositional logic, we cannot describe statements in terms of their properties or logical relationships.

Wumpus World PEAS description



Wumpus World PEAS description

- **Performance measure**
 - gold +1000, death -1000
 - -1 per step, -10 for using the arrow
- **Environment**
 - Squares adjacent to wumpus are smelly
 - Squares adjacent to pit are breezy
 - Glitter iff gold is in the same square
 - Shooting kills wumpus if you are facing it
 - Shooting uses up the only arrow
 - Grabbing picks up gold if in same square
 - Releasing drops the gold in same square
- **Sensors:** Stench, Breeze, Glitter, Bump, Scream
- **Actuators:** Left turn, Right turn, Forward, Grab, Release, Shoot

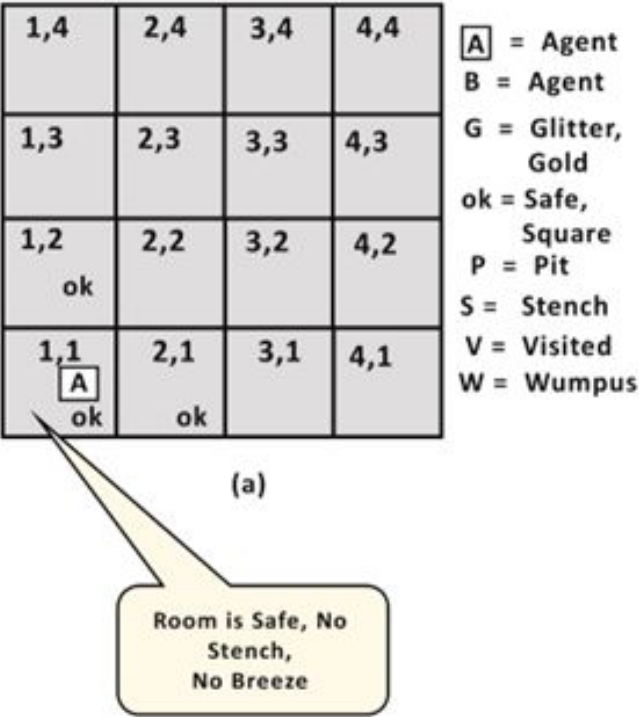
Sensors

- The agent will perceive the **stench** if he is in the room adjacent to the Wumpus. (Not diagonally).
- The agent will perceive **breeze** if he is in the room directly adjacent to the Pit.
- The agent will perceive the **glitter** in the room where the gold is present.
- The agent will perceive the **bump** if he walks into a wall.
- When the Wumpus is shot, it emits a horrible **scream** which can be perceived anywhere in the cave.
- These percepts can be represented as five element list, in which we will have different indicators for each sensor.
- Example if agent perceives stench, breeze, but no glitter, no bump, and no scream then it can be represented as:
[Stench, Breeze, None, None, None].

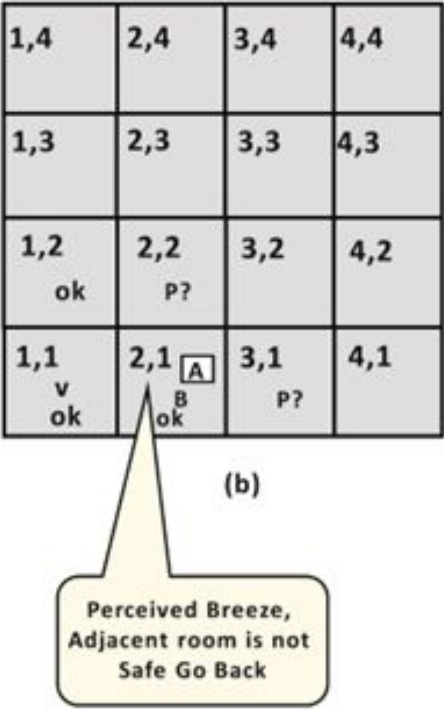
Wumpus World PEAS description

- **Partially observable:** The Wumpus world is partially observable because the agent can only perceive the close environment such as an adjacent room.
- **Deterministic:** It is deterministic, as the result and outcome of the world are already known.
- **Sequential:** The order is important, so it is sequential.
- **Static:** It is static as Wumpus and Pits are not moving.
- **Discrete:** The environment is discrete.
- **One agent:** The environment is a single agent as we have one agent only and Wumpus is not considered as an agent.

Agent's First step:



Second First step:



Agent's Third step:

1,4	2,4	3,4	4,4
1,3 w	2,3	3,3	4,3
1,2 A ok	2,2 P?	3,2	4,2
1,1 v ok	2,1 B	3,1 P?	4,1

A = Agent
 B = Agent
 G = Glitter, Gold
 ok = Safe,
 P = Pit
 S = Stench
 V = Visited
 W = Wumpus

(a)

Perceived
 stench ,
 No Breeze

Agent's Fourth step:

1,4	2,4 P?	3,4	4,4
1,3 W?	2,3 A S G B	3,3 P?	4,3
1,2 S V ok	2,2 V P?	3,2	4,2
1,1 v ok	2,1 B V ok	3,1 P?	4,1

(b)

Found gold

1,4	2,4 P?	3,4	4,4
1,3 W?	2,3 S G B	3,3	4,3
1,2	2,2 V P?	3,2	4,2
1,1 A ok	2,1 B V ok	3,1 P?	4,1

Atomic proposition variable for Wumpus world:

- Let $P_{i,j}$ be true if there is a Pit in the room $[i, j]$.
- Let $B_{i,j}$ be true if agent perceives breeze in $[i, j]$, (dead or alive).
- Let $W_{i,j}$ be true if there is wumpus in the square $[i, j]$.
- Let $S_{i,j}$ be true if agent perceives stench in the square $[i, j]$.
- Let $V_{i,j}$ be true if that square $[i, j]$ is visited.
- Let $G_{i,j}$ be true if there is gold (and glitter) in the square $[i, j]$.
- Let $OK_{i,j}$ be true if the room is safe.

Some Propositional Rules for the wumpus world:

$$(R1) \neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$$

$$(R2) \neg S_{21} \rightarrow \neg W_{11} \wedge \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$$

$$(R3) \neg S_{12} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{22} \wedge \neg W_{13}$$

$$(R4) S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$$

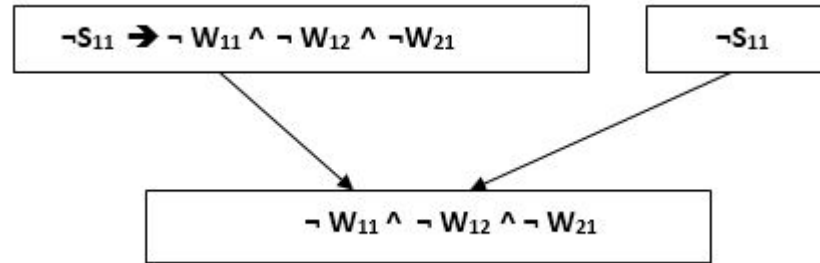
Following is the Simple KB for wumpus world when an agent moves from room [1, 1], to room [2,1]:

$\neg W_{11}$	$\neg S_{11}$	$\neg P_{11}$	$\neg B_{11}$	$\neg G_{11}$	V_{11}	OK_{11}
$\neg W_{12}$	----	$\neg P_{12}$	-----	----	$\neg V_{12}$	OK_{12}
$\neg W_{21}$	$\neg S_{21}$	$\neg P_{21}$	B_{21}	$\neg G_{21}$	V_{21}	OK_{21}

Prove that Wumpus is in the room (1, 3)

• **Apply Modus Ponens with $\neg S_{11}$ and R1:**

We will firstly apply MP rule with R1 which is $\neg S_{11} \rightarrow \neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, and $\neg S_{11}$ which will give the output $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$.

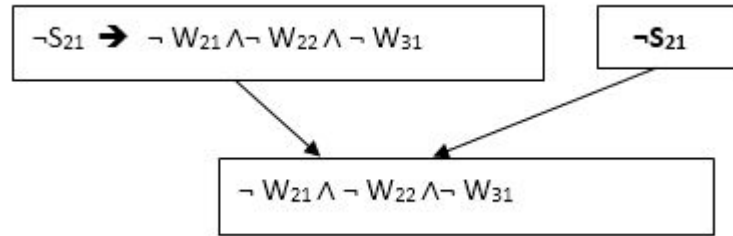


• **Apply And-Elimination Rule:**

After applying And-elimination rule to $\neg W_{11} \wedge \neg W_{12} \wedge \neg W_{21}$, we will get three statements: $\neg W_{11}$, $\neg W_{12}$, and $\neg W_{21}$.

• **Apply Modus Ponens to $\neg S_{21}$, and R2:**

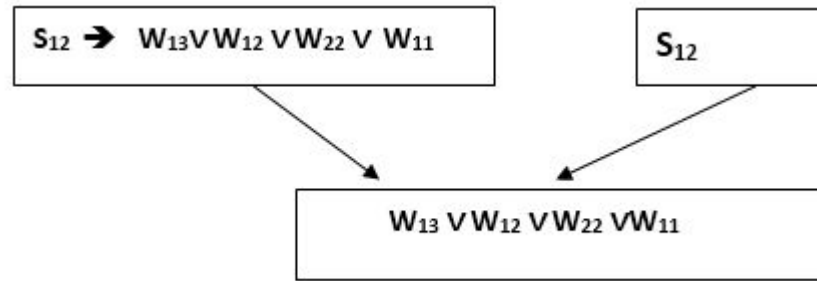
Now we will apply Modus Ponens to $\neg S_{21}$ and R2 which is $\neg S_{21} \rightarrow \neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$, which will give the Output as $\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$



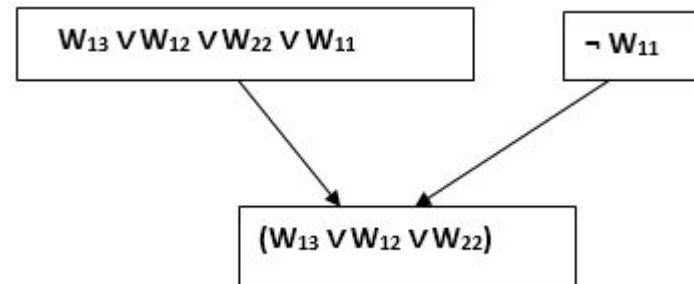
• **Apply And -Elimination rule:**

Now again apply And-elimination rule to $\neg W_{21} \wedge \neg W_{22} \wedge \neg W_{31}$, We will get three statements: $\neg W_{21}$, $\neg W_{22}$, and $\neg W_{31}$.

- **Apply MP to S_{12} and R4:**
- Apply Modus Ponens to S_{12} and R_4 which is $S_{12} \rightarrow W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$, we will get the output as $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$.

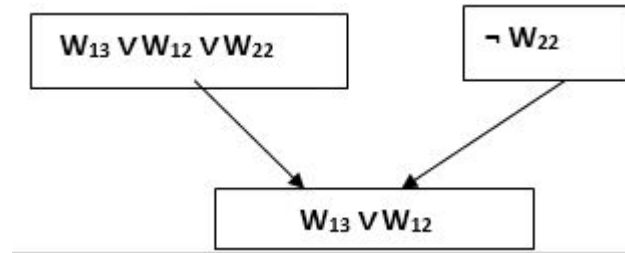


- **Apply Unit resolution on $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$ and $\neg W_{11}$:**
After applying Unit resolution formula on $W_{13} \vee W_{12} \vee W_{22} \vee W_{11}$ and $\neg W_{11}$ we will get $W_{13} \vee W_{12} \vee W_{22}$.



- **Apply Unit resolution on $W_{13} \vee W_{12} \vee W_{22}$ and $\neg W_{22}$:**

After applying Unit resolution on $W_{13} \vee W_{12} \vee W_{22}$, and $\neg W_{22}$, we will get $W_{13} \vee W_{12}$ as output.



Apply Unit Resolution on $W_{13} \vee W_{12}$ and $\neg W_{12}$

After Applying Unit resolution on $W_{13} \vee W_{12}$ and $\neg W_{12}$, we will get W_{13} as an output, hence it is proved that the Wumpus is in the room [1, 3].

